

Experience

Independent Research Engineer & Open-Source Developer

Feb 2026 - Present

a9l.im

- Perform independent interpretability and alignment research from hypothesis to experiments and write-up; publish tools and reproducible evaluation harnesses for open-weight LLMs.

Selected Research

Saklas

Apr 2026 - Present

Mechanistic interpretability workbench - Python, PyTorch

- Built a local workbench for extracting, probing, and steering LLMs with behavioral manifolds, with Mahalanobis whitening, support for Jacobian-lens readouts, and role-agnostic chat templates.
- Published a typed Python API, dashboard, and OpenAI/Ollama-compatible server tested on Gemma, Qwen, Mistral, Llama, GLM, gpt-oss, and Talkie. Available on PyPI with CUDA and MPS support.

Introspection via Kaomoji

Apr 2026 - Present

Representation study and Llmoji corpus - Python, Hugging Face

- Across five open-weight model families, classifying the opening kaomoji could predict the emotional content of a prompt with 80.6% accuracy on Gemma versus 11.1% by chance.
- Procrustes-aligned emotional category centroids to recover shared geometry; built kaomoji collection tool and published the corpus.

Elapsed-Time Encoding in LLMs

Jun 2026 - Present

Residual-stream probing study - Python, PyTorch

- Showed elapsed conversational time is linearly encoded at an elicited token ($r = 0.88$, about 0.3 seconds per token) by using linear probes.

Neuralese as a Conlang

Jul 2026 - Present

Ongoing Abstract-CoT analysis study - Python, PyTorch

- Implemented Abstract-CoT to test if a reasoning code could be interpreted as a compositional language.
- A preliminary 135M run did not successfully learn to use abstract tokens to reason.

Block Sparse Crosscoders

Jul 2026 - Present

Ongoing cross-layer neural geometry study - Python, PyTorch

- Synthesized recent work on block sparse autoencoders and sparse crosscoders to extract cross-layer multidimensional features.

Selected Engineering

- Agent infrastructure.** Gaslamp, a cross-review CLI for Claude Code and Codex, and Dagwood, a version-controlled DAG task tracker with a MCP.
- Interactive systems.** Built eight browser projects spanning relativity, MHD, nuclear engineering, options pricing, metabolism, epidemiology, redistricting, and comparative religion using custom web infrastructure.

Education

University of California, San Diego

Sep 2023 - Mar 2026

B.S. Mathematics - GPA 3.75 - completed early

Technical Skills

Research/ML: mechanistic interpretability; Python, PyTorch, NumPy/SciPy, scikit-learn, CUDA/MPS.**Engineering:** Java, JavaScript, WebGPU, Rust, FastAPI, Cloudflare Workers, MCP tooling, Git.